The $\mathcal{N}_{\rm esc}$ Recipe: A Cross-Regime Observation of Bekenstein-Bound Saturation from the Static Escrow Construction $|U|/T$

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Abstract

We propose a static escrow framework for gravitational entropy in which the basic ratio $S_{\rm esc} = |U|/T$ — gravitational binding energy divided by characteristic temperature — produces the Bekenstein-bound saturation form $\mathcal{N}_{\rm esc}(E, L) \equiv 2\pi EL/(\hbar c)$ across three distinct regimes: (i) a test mass in Schwarzschild geometry at the horizon limit, (ii) the Bekenstein-Hawking entropy of a black-hole horizon via the Smarr formula, and (iii) the entanglement-entropy change of a localized matter configuration in a Rindler wedge. The dimensionless count $\mathcal{N}_{\rm esc}(E, L)$ is identical in form to the Bekenstein-bound saturation value; what the framework names is the *recipe* by which (E, L) are extracted from $|U|/T$ in each gravitational regime. The same one-function recipe, applied across three qualitatively distinct settings, evaluates to the Bekenstein form in each. For regime (iii), we identify the framework's prediction with Casini's quantum-field-theoretic derivation of the Bekenstein bound. We provide first-principles lattice verification of the Rindler-wedge inequality across mass-perturbation and Compton-wavepacket protocols, lattice sizes $N \in [200, 1200]$, and perturbation strengths $m^2_{\rm pert} \in [0.5, 5.0]$ spanning one order of magnitude. The maximum observed saturation is 5.4% at the Compton scale. The framework is an organizing observation, not a derivation: each regime's prediction follows from a standard continuum result (Bisognano-Wichmann theorem, Smarr formula, Bekenstein bound), while the framework's contribution is the unifying observation that the same recipe $|U|/T$ generates the Bekenstein form in all three. The present work is the fifth deposit in the Windstorm Institute gravitational entropy escrow series [15–19] and formalizes the cross-regime $\mathcal{N}_{\rm esc}$ notation introduced in [19].

1 Introduction

The Bekenstein bound [1] asserts that the entropy contained in any region of size R with total energy E is bounded by:

$$S \leq \frac{2\pi ER}{\hbar c}. \tag{1}$$

This bound, originally motivated by black-hole thermodynamics, has since been derived from quantum field theory by Casini [2], who showed that for a localized configuration in a Rindler wedge, the entanglement-entropy change $\Delta S_{\rm EE}$ satisfies precisely the form of Eq. 1.

In parallel, the Bisognano-Wichmann theorem [3,4] identifies the vacuum modular Hamiltonian of a half-space cut in Minkowski space with the boost generator, providing the structural backbone for understanding entanglement in relativistic field theory. The Smarr formula [5] expresses the mass of a stationary black hole as a sum of horizon contributions weighted by intensive thermodynamic variables.

The present work observes that these three results — Bekenstein bound for matter, Bekenstein-Hawking entropy from Smarr, and the BW-Casini bound for entanglement — share a common structural form: the ratio of energy to temperature, $S = E/T$, where T is the characteristic temperature of the relevant horizon (Hawking, Unruh) and E is the total energy of the configuration.

We propose to call this the *static escrow ratio*:

$$S_{\text{esc}} = \frac{|U|}{T}, \tag{2}$$

where "escrow" is a deliberately neutral label for the dimensionless entropy-capacity quantity extracted by evaluating the ratio $|U|/T$ within a regime-specific thermodynamic sector; we do not attach any dynamical or ontological commitment to the term in the present work. We examine the behavior of S_{esc} across three regimes. In each, S_{esc} takes the Bekenstein-bound saturation form $2\pi ER/(\hbar c)$ with appropriate identifications. The static escrow postulate was introduced as a physical reframing of gravitational binding energy in [16], where it was used to organize Newton's law, Bekenstein-Hawking entropy, the equivalence principle, and the deep-MOND Tully-Fisher relation under a single bookkeeping picture. The present work isolates a specific structural consequence of that postulate — the cross-regime appearance of the Bekenstein form — and formalizes it as the function-and-recipe pair described in §1.1.

Relation to prior work in this series. The framework's empirical content has been examined in four prior preprints. [18] reports a direct lattice quantum field theory test of the literal bipartition-entropy reading of the postulate (falsified in both 1+1D and 3+1D) and of the modular-Hamiltonian reading (partial survival in 1+1D, with a small-d_1 window producing a prefactor $\approx 1/30$ of the literal Bisognano-Wichmann value). [17] examines a candidate covariant extension that turned out to be algebraically identical to a 1981 Bekenstein result. [15] derives a non-equilibrium efficiency bound for Bose-Einstein condensate analog gravity from a Verlinde-based construction in the same thermodynamic family. [19] translates four standard general-relativistic results (gravitational time dilation, Tolman temperature law [14], Bekenstein-Hawking entropy [12], Jacobson's thermodynamic derivation of Einstein's equations [13]) into the escrow vocabulary, and introduces the working notation $\mathcal{N}_{\text{esc}}$ for the dimensionless escrow count in the test-mass and black-hole sectors. The present work refines that notation by formalizing $\mathcal{N}_{\text{esc}}$ as a single two-argument function (§1.1) and extending the cross-regime structure to the localized-perturbation sector via Casini's bound.

1.1 The $\mathcal{N}_{\text{esc}}$ function and the escrow recipe

When $S_{\text{esc}} = |U|/T$ is reduced to dimensionless form, every regime examined in this work produces a value of the same two-argument function:

$$\boxed{\mathcal{N}_{\text{esc}}(E, L) \equiv \frac{2\pi\, E\, L}{\hbar c}} \tag{3}$$

where E has units of energy and L has units of length. We call $\mathcal{N}_{\text{esc}}$ the *escrow count*. It is dimensionless and is identical in form to the Bekenstein-bound saturation value $S_{\text{Bek}}(E, L)$.

What is named, and what is not. The numerical value of $\mathcal{N}_{\text{esc}}(E, L)$ has been known since [1]; we make no claim of priority for the combination $2\pi EL/(\hbar c)$. (A separate methodological lesson on the importance of checking proposed escrow extensions against the 1981 Bekenstein literature is documented in [17], where a candidate covariant extension was shown to be algebraically identical to a 1981 Bekenstein result.) What the escrow framework names is the *recipe* by which (E, L) are extracted in each gravitational regime — namely, the static escrow construction

$$S_{\mathrm{esc}} = \frac{|U|}{T} \;\longrightarrow\; (E, L)_{\mathrm{regime}} \;\longrightarrow\; \mathcal{N}_{\mathrm{esc}}(E, L), \tag{4}$$

where $|U|$ is the regime's gravitational binding energy and T is the relevant horizon (Hawking, Unruh) temperature. The two arrows denote the two-step recipe: first, the regime's $|U|/T$ construction is decomposed into an energy E and a length L (regime-specific identification); second, the Bekenstein-form function $\mathcal{N}_{\mathrm{esc}}(E, L) = 2\pi EL/(\hbar c)$ is evaluated at those arguments. The framework's content is the observation that this recipe, applied across qualitatively distinct gravitational regimes, always lands on the Bekenstein-bound saturation value $\mathcal{N}_{\mathrm{esc}}(E, L)$. The function is Bekenstein's; the recipe is the framework's.

The three regime evaluations. Each of the regimes treated in this work corresponds to a specific (E, L) at which the escrow function is evaluated by the recipe in Eq. 4:

$$\mathcal{N}_{\mathrm{esc}}^{(\mathrm{II})} = \mathcal{N}_{\mathrm{esc}}\big(mc^2,\, r_s\big) = \frac{2\pi r_s}{\bar{\lambda}_C}, \tag{5}$$

$$\mathcal{N}_{\mathrm{esc}}^{(\mathrm{III})} = \mathcal{N}_{\mathrm{esc}}\big(Mc^2,\, r_s\big) = \frac{A}{4\ell_P^2} = 4\pi\left(\frac{M}{m_P}\right)^2, \tag{6}$$

$$\mathcal{N}_{\mathrm{esc}}^{(\mathrm{IV})} = \mathcal{N}_{\mathrm{esc}}(\Delta E_A,\, d) = \frac{2\pi\,\Delta E_A\, d}{\hbar c}. \tag{7}$$

Here $\bar{\lambda}_C = \hbar/(mc)$ is the test mass's reduced Compton wavelength, $m_P = \sqrt{\hbar c/G}$ is the Planck mass, $\ell_P = \sqrt{\hbar G/c^3}$ is the Planck length, and ΔE_A, d are the localized-perturbation energy and characteristic distance of §4. Equations 5–7 are three evaluations of a single function at regime-specific arguments — *not* three independently defined quantities sharing a family resemblance.

The Smarr partition lives in the recipe, not the function arguments. In the two horizon regimes (§2 and §3), the escrow ratio computed from $|U|/T_H$ contains an internal factor of 1/2 arising from the Smarr-formula homogeneity of Schwarzschild thermodynamics: $S_{\mathrm{esc}}^{(\mathrm{II})} = (mc^2/2)/T_H$ and $S_{\mathrm{esc}}^{(\mathrm{III})} = (Mc^2/2)/T_H$, where the rest energy is partitioned because M is homogeneous of degree 1/2 in the horizon area (treated in §3). The 1/2 in the numerator combines with the 4π in $1/T_H = 4\pi r_s/(\hbar c)$ to produce the 2π prefactor of the function $\mathcal{N}_{\mathrm{esc}}$ evaluated at the full rest energy and length r_s. The partition is therefore a property of the *recipe* (the way the escrow construction extracts $|U|/T$ in Smarr-homogeneous regimes) and not of the function arguments themselves. For the localized-perturbation regime §4, no such partition appears because there is no Smarr-type area homogeneity to invoke; the recipe extracts the full perturbation energy ΔE_A and the characteristic distance d directly.

Interpretation. The dimensionless count $\mathcal{N}_{\mathrm{esc}}(E, L)$ admits the standard Bekenstein-bound reading: it is the upper bound on the entropy (in units of k_B) of a system of energy E confined to a region of characteristic size L. Equivalently, it measures the action EL/c of the configuration in units of $\hbar/(2\pi)$. The framework's claim is not that this number is novel — it is well-known — but that the escrow recipe $|U|/T$ produces *this particular* dimensionless count in every gravitational regime to which it has been applied.

1.2 Scope and limitations

We are explicit about what this work does and does not provide:

- **What this work provides:** A cross-regime organizing observation that the static escrow ratio $|U|/T$ produces the Bekenstein-bound form in three distinct gravitational settings, with empirical anchoring of the Rindler-wedge sector by first-principles lattice computation reproducing the continuum BW identity within lattice-discretization precision ($\sim 0.1\%$), and with the inequality saturation level (maximum 5.4% at Compton scale) characterized across parameter space.

- **What this work does not provide:** A derivation of any individual regime's prediction from the postulate $S_{\text{esc}} = |U|/T$ alone. Each regime's prediction follows from a standard continuum result. The framework's value is the unifying observation, not a new derivation.

2 The Test-Mass Sector

Consider a test mass m at radial coordinate r in Schwarzschild geometry with Schwarzschild radius $r_s = 2GM/c^2$. Let $f(r) = 1 - r_s/r = 1 - 2GM/(rc^2)$ be the metric function. The gravitational binding energy of the test mass relative to spatial infinity is:

$$|U_{\text{grav}}(r)| = mc^2 \left(1 - \sqrt{f(r)}\right), \tag{8}$$

which approaches mc^2 as $r \to r_s$. The Hawking temperature [12] of the horizon, observed at infinity, is:

$$T_H = \frac{\hbar c^3}{8\pi G M k_B} = \frac{\hbar c}{4\pi r_s k_B}. \tag{9}$$

The Smarr partition. For a Schwarzschild black hole, the Smarr formula $Mc^2 = 2T_H S_{BH}$ (treated more fully in §3) implies that a mass deposit δM onto the horizon contributes entropy

$$\delta S_{BH} = \frac{\delta M \cdot c^2/2}{T_H}, \tag{10}$$

i.e., only half of the rest energy enters the thermodynamic ratio. We adopt this Smarr-partitioned convention for the test-mass escrow ratio in the horizon limit, paralleling the partition used in §3. We acknowledge that the extension from the horizon (where the Smarr partition is derived from Euler's theorem applied to the Schwarzschild mass functional) to a test mass at the horizon limit is heuristic rather than independently derived: a test particle of rest energy mc^2 falling into a Schwarzschild black hole deposits entropy $\delta S_{BH} = mc^2/(2T_H)$ onto the horizon, and we interpret this expression as also characterizing the escrow ratio of the test mass itself at $r \to r_s$. The structural significance of this choice is not the factor itself, but that the same Smarr partition appearing naturally in horizon thermodynamics also organizes the test-mass horizon-limit expression into Bekenstein form.

Forming the escrow ratio in the limit $r \to r_s$ (test mass approaching the horizon) with the Smarr partition:

$$S_{\text{esc}}^{(\text{II})} = \frac{|U_{\text{grav}}|/2}{T_H} \xrightarrow{r \to r_s} \frac{mc^2/2}{T_H} = \frac{2\pi r_s mc}{\hbar} = \frac{2\pi r_s}{\bar{\lambda}_C}, \tag{11}$$

where $\bar{\lambda}_C \equiv \hbar/(mc) = \lambda_C/(2\pi)$ is the reduced Compton wavelength of the test mass and $\lambda_C = h/(mc)$ is the standard Compton wavelength. In dimensionless form:

$$\mathcal{N}_{\text{esc}}^{(\text{II})} = \frac{2\pi r_s}{\bar{\lambda}_C} = \frac{2\pi r_s\, mc}{\hbar}, \tag{12}$$

which is precisely the Bekenstein bound for a system of energy mc^2 confined within radius r_s:

$$\mathcal{N}_{\rm esc}^{\rm (II)} = \frac{2\pi\,(mc^2)\,r_s}{\hbar c} = S_{\rm Bek}(E = mc^2,\ R = r_s). \tag{13}$$

Remark 1. *The Smarr partition is essential to the structural parallel and lives in the* recipe *(the escrow ratio $S_{\rm esc}^{\rm (II)} = (|U|/2)/T_H$), not in the function arguments. Without it, the unpartitioned ratio mc^2/T_H would overcount by a factor of two relative to the Bekenstein bound, just as Mc^2/T_H would overcount S_{BH} by two without the Smarr partition in §3. The $1/2$ in the recipe's numerator combines with the 4π in $1/T_H$ to produce the 2π prefactor of the function $\mathcal{N}_{\rm esc}$ evaluated at the full rest energy. The factor of two is not a fudge; it is the same homogeneity-of-mass-in-area structure that produces Smarr's relation.*

Remark 2. *This identification is evaluated specifically at the horizon limit $r \to r_s$ using T_H. At intermediate radii $r > r_s$, an alternative formulation using the local Unruh temperature $T_U(r)$ recovers a Newtonian Bekenstein-form $\mathcal{N}_{\rm esc} \propto 2\pi r/\bar{\lambda}_C$; however, $T_U(r)$ diverges in the $r \to r_s$ limit, so a smooth interpolation between the local-Unruh formulation at $r > r_s$ and the Hawking-temperature identification at $r = r_s$ requires explicit Tolman redshifting [14] between the two reference frames. We defer the unified $T_U \to T_H$ smooth-interpolation treatment to future work; the horizon-limit identification suffices for the cross-regime structural observation of this paper.*

This is a structural identity check: the Smarr-partitioned escrow ratio reproduces the Bekenstein bound for matter at the horizon-saturation scale.

3 The Black-Hole Horizon Sector

For a stationary, uncharged, non-rotating black hole, the Smarr formula [5] gives:

$$Mc^2 = 2T_H S_{BH}, \tag{14}$$

where $S_{BH} = A/(4\ell_P^2)$ is the Bekenstein-Hawking entropy with ℓ_P the Planck length and A the horizon area. Equation 14 is a homogeneity statement: M is homogeneous of degree $1/2$ in the area A for a Schwarzschild horizon, and Euler's theorem yields $Mc^2 = 2(\partial Mc^2/\partial A)A = 2T_H S_{BH}$.

Identifying $|U_{\rm grav}| = Mc^2/2$ (the Smarr-formula partition of total mass into horizon contribution) and $T = T_H$:

$$S_{\rm esc}^{\rm (III)} = \frac{Mc^2/2}{T_H} = S_{BH} = \frac{A}{4\ell_P^2}. \tag{15}$$

In dimensionless form, the recipe's result is the evaluation $\mathcal{N}_{\rm esc}(Mc^2, r_s)$ of the escrow function of §1.1 — with the $1/2$ from the Smarr partition residing in the recipe (the $Mc^2/2$ numerator of $S_{\rm esc}^{\rm (III)}$ in Eq. 15) rather than in the function argument:

$$\mathcal{N}_{\rm esc}^{\rm (III)} = \mathcal{N}_{\rm esc}\big(Mc^2, r_s\big) = \frac{A}{4\ell_P^2} = \frac{4\pi G M^2}{\hbar c} = 4\pi\left(\frac{M}{m_P}\right)^2, \tag{16}$$

where $m_P = \sqrt{\hbar c/G}$ is the Planck mass. The escrow ratio reproduces the Bekenstein-Hawking entropy exactly. This is a recovery check: the framework is consistent with the known thermodynamic structure of black holes, and the same $|U|/2$ partition in the escrow ratio used in §2 arises here from Smarr's relation directly.

4 The Localized-Perturbation Sector

This is the regime in which the framework's content can be tested by direct computation. We consider a localized energy perturbation of a Minkowski vacuum, with the entangling surface taken to be a half-space cut in a Rindler wedge.

4.1 The escrow prediction

For a Rindler observer at proper distance d from the cut, the Unruh temperature is:

$$T_U = \frac{\hbar c}{2\pi d k_B}. \tag{17}$$

For a localized configuration of total in-region energy ΔE_A at characteristic distance d:

$$S_{\text{esc}}^{(\text{IV})} = \frac{\Delta E_A}{T_U} = \frac{2\pi\, \Delta E_A\, d}{\hbar c}. \tag{18}$$

This is the evaluation $\mathcal{N}_{\text{esc}}(\Delta E_A, d)$ of the escrow function of §1.1:

$$\mathcal{N}_{\text{esc}}^{(\text{IV})} = \mathcal{N}_{\text{esc}}(\Delta E_A, d) = \frac{2\pi\, \Delta E_A\, d}{\hbar c}. \tag{19}$$

We will show this is precisely the Bekenstein bound on ΔS_{EE} for the configuration, derivable from BW theorem and the first law of entanglement.

4.2 Derivation of the Escrow Form of the Casini–BW Inequality

Theorem 1 (Escrow Form of the Casini–BW Inequality, conditional on moment-positivity)**.** *Let ρ be a state of a free scalar quantum field theory that differs from the Minkowski vacuum by a localized energy perturbation, with energy density $\Delta T_{00}(x) = \langle T_{00}(x)\rangle_\rho - \langle T_{00}(x)\rangle_{\text{vac}}$ supported within distance d from the entangling surface of a half-space Rindler bipartition. Let $\Delta E_A = \int_A \Delta T_{00}(x)\, dx$ be the total in-region energy. Suppose the configuration satisfies the moment-bound condition*

$$\int_A x\, \Delta T_{00}(x)\, dx \le d \cdot \Delta E_A \quad \textit{(moment-positivity assumption).} \tag{20}$$

Then

$$\Delta S_{\text{EE}} \le \frac{2\pi\, \Delta E_A\, d}{\hbar c}. \tag{21}$$

The moment-bound condition holds rigorously when $\Delta T_{00}(x) \ge 0$ pointwise. For QFT states where vacuum polarization can produce regions of negative ΔT_{00}, the condition is not guaranteed in general; we verify it empirically in Section 4.3 for the protocols studied here (Table 4, ratios 0.98–0.999 *across parameter combinations).*

Proof. The derivation proceeds in three steps.

Step 1 (BW identification). For a half-space Rindler bipartition of Minkowski vacuum, the Bisognano-Wichmann theorem identifies the vacuum modular Hamiltonian with the boost generator:

$$K_A = 2\pi \int_{x\in A} x\, T_{00}(x)\, dx. \tag{22}$$

For any state ρ, the change in modular-Hamiltonian expectation is:

$$\Delta\langle K_A\rangle = 2\pi \int_A x\, \Delta T_{00}(x)\, dx. \tag{23}$$

Step 2 (First law of entanglement). The Wall identity [6] decomposes the modular Hamiltonian change as:

$$\Delta\langle K_A\rangle = \Delta S_{\mathrm{EE}} + S_{\mathrm{rel}}(\rho \,\|\, \rho_{\mathrm{vac}}), \tag{24}$$

where $S_{\mathrm{rel}}(\rho \,\|\, \rho_{\mathrm{vac}}) \geq 0$ by Klein's inequality. Therefore:

$$\Delta S_{\mathrm{EE}} \leq \Delta\langle K_A\rangle. \tag{25}$$

Step 3 (Moment bound). By the moment-positivity assumption (Eq. 20),

$$\int_A x\,\Delta T_{00}(x)\,dx \leq d \cdot \int_A \Delta T_{00}(x)\,dx = d \cdot \Delta E_A. \tag{26}$$

Combining Steps 1 and 3:

$$\Delta\langle K_A\rangle = 2\pi \int_A x\,\Delta T_{00}(x)\,dx \leq 2\pi\, d\,\Delta E_A. \tag{27}$$

Chaining Eq. 25 and Eq. 27:

$$\Delta S_{\mathrm{EE}} \leq \Delta\langle K_A\rangle \leq \frac{2\pi\,\Delta E_A\, d}{\hbar c}. \tag{28}$$

This is the Escrow Inequality. □

Remark 3. *Theorem 1 reproduces Casini's QFT derivation of the Bekenstein bound [2], with the escrow postulate $S_{\mathrm{esc}} = |U|/T$ identifying the right-hand-side bound. The two contributions are complementary: Casini provides the proof; the escrow framework provides the cross-regime interpretation.*

Remark 4. *The boost-generator expectation $\Delta\langle K_A^{\mathrm{boost}}\rangle = 2\pi \int_A x\Delta T_{00}\,dx$ is an* equality, *not an inequality, by direct evaluation. For point perturbations at distance d, this evaluates exactly to $2\pi\, d\,\Delta E_A$. The inequality emerges only when this is converted to a bound on ΔS_{EE} via the first law of entanglement, which introduces the (always non-negative) relative entropy.*

4.3 Lattice Verification

We verify the Escrow Inequality on a 1+1-dimensional lattice scalar field theory. The lattice methodology — free-scalar discretization, Gaussian-state covariance machinery, Williamson decomposition for symplectic eigenvalues, and Bisognano-Wichmann identification of the boost generator with the modular Hamiltonian — follows the protocol of [18], which examined the literal bipartition-entropy reading of the escrow postulate (ruled out by 10^{56} orders of magnitude across a 295-point parameter grid) and the modular-Hamiltonian reading (partial survival in 1+1D within a small-d_1 window). The present work tests the bound on ΔS_{EE} given by Theorem 1, which is the surviving content of the framework's Rindler-wedge prediction in the inequality-on-ΔS_{EE} form.

The lattice discretization is:

$$H = \frac{1}{2}\sum_i \left[\pi_i^2 + (\phi_{i+1} - \phi_i)^2 + m_i^2\phi_i^2\right], \tag{29}$$

where the sum runs over N sites with Dirichlet boundary conditions and m_i^2 is the (possibly site-dependent) mass-squared. This is the standard discretized harmonic chain whose entanglement properties have been studied extensively in [9]; for the general methodology of entanglement-entropy computation in free quantum field theory we follow the framework of [10]. The Gaussian vacuum is constructed by diagonalizing the resulting quadratic Hamiltonian; entanglement entropies and modular Hamiltonians are computed via Williamson decomposition of reduced covariance matrices.

Two protocols are examined:

Protocol A: Compton wavepacket. Set $m_i^2 = m^2$ uniformly and prepare a one-particle Gaussian wavepacket of Compton width $\sigma_c = 1/m$ centered at distance d_1 from the cut. The perturbed covariance is computed analytically from the wavepacket mode.

Protocol B: Mass perturbation. Set $m_i^2 = m_{\text{pert}}^2 \delta_{i,d_1}$ (single-site mass insertion at distance d_1 from the cut, on a massless background). The ground state is recomputed in the perturbed Hamiltonian.

4.3.1 Wavepacket protocol results

For Protocol A with $m = 0.05$, $\sigma_c = 20$, $N = 1200$:

d_1	ΔS_{EE}	ΔE_A	$2\pi d_1 \Delta E_A$	ΔS_{EE}/bound
20	1.228	0.045	22.70	0.054
40	1.356	0.053	33.58	0.040
80	1.386	0.055	48.78	0.028
100	1.386	0.056	55.76	0.025
200	1.386	0.056	90.61	0.015
300	1.386	0.056	125.46	0.011

Table 1: Escrow Inequality verification for Compton-wavepacket protocol. Maximum saturation 5.4% at $d_1 \approx \sigma_c$ (Compton scale). The bound is loose and becomes looser at larger d_1.

The saturated value $\Delta S_{\text{EE}} \to 2\ln 2$ for $d_1 \gg \sigma_c$ is the well-known single-mode bosonic Gaussian entropy at occupation $\langle n \rangle = 1$ [7, 8].

4.3.2 Mass-perturbation protocol results

For Protocol B with $m_{\text{pert}}^2 = 1$ at single site d_1, $N = 800$:

d_1	ΔS_{EE}	ΔE_A	$2\pi d_1 \Delta E_A$	ratio
5	−0.639	0.257	8.07	−0.079
10	−0.532	0.257	16.12	−0.033
20	−0.421	0.257	32.23	−0.013
40	−0.309	0.256	64.45	−0.005
80	−0.200	0.256	128.87	−0.002
160	−0.100	0.256	257.58	−0.0004

Table 2: Escrow Inequality verification for mass-perturbation protocol. ΔS_{EE} is negative due to vacuum rearrangement, so the bound $\Delta S_{\text{EE}} \leq 2\pi d_1 \Delta E_A/(\hbar c)$ is trivially satisfied with the entire positive RHS as slack. This protocol therefore does not provide a tight verification of the inequality; only Table 1 (the Compton-wavepacket protocol, maximum 5.4% saturation) provides a non-trivial test.

4.3.3 Boost-generator equality verification

To verify the BW identification (Step 1 of Theorem 1), we compare the modular Hamiltonian expectation computed two ways: via the lattice modular Hamiltonian (Williamson decomposition) versus via direct evaluation of the boost generator $K_A^{\text{boost}} = 2\pi \sum_{i \in A} (i - \text{cut}) \cdot T_{00}(i)$.

N	d_1	m^2_{pert}	$\Delta\langle K_A^{\text{boost}}\rangle/(2\pi d_1 \Delta E_A)$
200	40	1.0	0.99884
200	80	1.0	0.99899
400	80	1.0	0.99937
400	160	1.0	0.99943
800	80	1.0	0.99938
800	160	1.0	0.99967
800	320	1.0	0.99969
1200	40	0.5	0.99585
1200	80	2.0	0.99992
1200	160	5.0	1.00013

Table 3: Boost-generator equality verification. Mean ratio 0.99913 ± 0.00122 across 10 parameter combinations. This validates Step 1 of Theorem 1 (BW identification of the boost generator with the continuum modular Hamiltonian).

4.3.4 Moment-inequality verification

For mass-perturbation Protocol B, we verify Step 3 of the proof (Eq. 26):

d_1	$\int_A x\Delta T_{00}$	$d_1 \cdot \int_A \Delta T_{00}$	ratio
5	1.261	1.284	0.982
10	2.544	2.566	0.991
20	5.108	5.130	0.996
40	10.235	10.258	0.998
80	20.485	20.510	0.999
160	40.967	40.994	0.999

Table 4: Moment-inequality verification. The ratio is consistently below 1, validating Eq. 26. The slight gap arises from negative-ΔT_{00} regions due to vacuum polarization, which reduce (rather than violate) the first moment.

4.4 Saturation conditions and structural observations

The Escrow Inequality is generally loose: saturation requires both $S_{\text{rel}} \to 0$ and saturation of the moment bound (Eq. 26). The first condition requires fine-tuned coherent states (the perturbed state must be modular-flow-aligned with the vacuum); the second requires all energy to be concentrated at the maximum extent d.

For generic Compton-localized states, both conditions fail by large margins. The maximum saturation observed in our protocols is 5.4%, occurring at the Compton scale $d_1 \approx \sigma_c$ in the wavepacket protocol.

5 Cross-Regime Synthesis

The three regimes are three evaluations of the single escrow function $\mathcal{N}_{\text{esc}}(E, L) = 2\pi EL/(\hbar c)$ of §1.1 at regime-specific arguments extracted by the recipe $|U|/T$:

The framework's content is the observation that the same one-function recipe, applied across three qualitatively distinct gravitational settings, always evaluates to the Bekenstein form $2\pi EL/(\hbar c)$. We make no claim to derive any individual prediction from the postulate

Regime	$\lvert U\rvert/T$ recipe	T	E	L	$\mathcal{N}_{\text{esc}}(E,L)$
§2 Test mass	$(mc^2/2)/T_H$	T_H	mc^2	r_s	$2\pi r_s/\bar{\lambda}_C$
§3 BH horizon	$(Mc^2/2)/T_H$	T_H	Mc^2	r_s	$4\pi(M/m_P)^2$
§4 Localized matter	$\Delta E_A/T_U$	T_U	ΔE_A	d	$2\pi\Delta E_A d/(\hbar c)$

Table 5: Cross-regime structural form of the escrow recipe. Column "$\lvert U\rvert/T$ recipe" shows the escrow construction in each regime, with the Smarr partition (rest energy)/2 residing in the numerator for §2 and §3; this partition is a property of the recipe (the way $\lvert U\rvert/T$ is extracted in Smarr-homogeneous regimes) rather than of the function. The function $\mathcal{N}_{\text{esc}}(E,L) = 2\pi EL/(\hbar c)$ — identical in form to the Bekenstein-bound saturation value — is then evaluated at the regime-specific energy E and length scale L. All three regimes land on the Bekenstein form. The function is not new; the recipe is the framework's contribution.

alone; each follows from a standard continuum result (Bekenstein 1981 for §2; Smarr 1973 for §3; Casini 2008 for §4). The unifying observation is the contribution.

6 Guardrails and Falsifiability

We are explicit about what the framework does *not* claim, and what would constitute falsification.

6.1 Retracted Claims

The development of this framework involved multiple iterations across earlier preprints. We explicitly retract the following claims that appeared in versions v0.6 and earlier and did not survive subsequent audit:

- A specific universal prefactor of 1/30 in the Rindler-wedge modular Hamiltonian content (this value, first reported in [18] as an approximate small-d_1 window observation in 1+1D, was subsequently determined to be a window-fitting artifact rather than a universal saturation prefactor — at larger d_1 the local exponent decreases smoothly into a sublinear decay tail).
- "Halving per spatial dimension" of the saturation prefactor (not reproducible under independent verification; depends on the specific perturbation protocol chosen).
- A direct equality $\Delta\langle K\rangle = 2\pi m d_1$ identified with rest mass m (the correct identification is with total perturbation energy ΔE_A, not rest mass; and the relation is an equality of the boost generator with $2\pi d\Delta E_A$, not of the lattice modular Hamiltonian).
- A claim that $\Delta S_{\text{EE}} = 2\ln 2$ per particle is a derivation from the escrow postulate (it is a textbook result of Gaussian quantum information [7, 8], consistent with the framework but not derived from it).
- A previous five-step derivation of an inequality $\Delta\langle K\rangle \le 2\pi E d$ which contained a logic gap in its final step (the corrected statement is the bound on ΔS_{EE} given in Theorem 1).

6.2 Falsifiability

The framework's empirical content can be falsified by:

1. **Inequality violation:** Discovery of any localized perturbation protocol producing $\Delta S_{\text{EE}} > 2\pi\Delta E_A d/(\hbar c)$ in free QFT. (Equivalent to falsifying Casini's bound.)

2. **Cross-regime form violation:** Discovery of a gravitational regime in which $|U|/T$ does not produce the Bekenstein-bound saturation form $2\pi ER/(\hbar c)$.

3. **Boost-generator equality violation:** Discovery of a parameter regime in which $\Delta\langle K_A^{\rm boost}\rangle/(2\pi d\Delta E_A)$ deviates substantially from unity in continuum BW theory.

None of these has been found in the present work or in cross-validation against other systems.

7 Discussion

7.1 What the framework is not, and what it does not derive

The framework is not a quantum theory of gravity, nor a derivation of the Bekenstein bound from a deeper principle, nor a source of new numerical predictions. Calling the cross-regime observation a "level jump" in physical understanding would overstate the contribution. Specifically:

- The Bekenstein bound itself is not derived from the escrow postulate; it follows from the BW theorem and the first law of entanglement [2].
- The Bekenstein-Hawking entropy is not derived from the escrow postulate; it follows from the Smarr formula [5].
- The specific values of $\Delta S_{\rm EE}$ for one-particle states ($2\ln 2$) and multi-particle states ($N\cdot 2\ln 2$) follow from standard Gaussian quantum information [7,8], not from the framework.

The framework's content is restricted to the observation that the same recipe $|U|/T$ produces the Bekenstein-saturation form $\mathcal{N}_{\rm esc}(E,L) = 2\pi EL/(\hbar c)$ across three regimes, providing a unifying interpretive structure.

7.2 Why the cross-regime recurrence is not algebraically trivial

A skeptical reader may ask whether the cross-regime appearance of the Bekenstein form is a tautology, on either of two grounds: (i) that all three regimes already *contain* the combination $2\pi EL/(\hbar c)$ by the way their underlying derivations are set up, so that any organizing variable would algebraically produce this form; or (ii) that the combination $EL/(\hbar c)$ is the unique dimensionless ratio constructible from $(E, L, \hbar, c)$, so the appearance of this form is dimensionally forced once one commits to organizing entropy by energy and length.

The second concern is partly valid. Given the inputs $(E, L, \hbar, c)$ and the requirement of a dimensionless output, $EL/(\hbar c)$ is indeed the unique combination; the factor 2π is conventional from [1]. The function $\mathcal{N}_{\rm esc}$ is therefore dimensionally inevitable once (E, L) have been chosen as the inputs — which is why the present work does not claim novelty for the function itself.

The framework's claim is narrower: that the static escrow construction $|U|/T$ — without invoking any auxiliary length or energy scale beyond what each gravitational regime supplies internally — extracts a specific (E, L) pair in each regime and that the resulting $\mathcal{N}_{\rm esc}(E, L)$ equals the regime's Bekenstein-form result with the correct prefactor. This is the content the dimensional argument does not deliver: it is not dimensional analysis but a specific structural recurrence under a specific recipe.

To address the first concern: the three derivations come from structurally distinct mathematical frameworks. The Bekenstein bound (§2) is an information-theoretic statement about entropy capacity in a region of given energy. The Smarr formula (§3) is a classical-thermodynamic homogeneity statement about M as a function of horizon area, derived from Euler's theorem applied to the Schwarzschild mass functional. The BW-Casini bound (§4) is a relative-entropy inequality on the half-space modular Hamiltonian of a free QFT. These are structurally distinct frameworks — capacity theory, classical thermodynamic homogeneity, and von Neumann algebraic modular theory — even though one structural link does exist (Casini derives the Bekenstein bound from BW theory, so the §2 and §4 sectors share a modular-theoretic foundation via Casini's 2008 result). The Smarr-vs-modular contrast, however, is genuine: the homogeneity-of-M-in-area argument is a classical thermodynamic identity independent of any modular structure, and its agreement with the modular-theoretic results under the recipe $|U|/T$ is not algebraically forced. The observation-dependence of entropy structures across these distinct frameworks has been previously emphasized in [11].

Uniqueness of the recipe is not established. We have not shown that $|U|/T$ is the *only* recipe that produces a cross-regime Bekenstein-form result; we have shown only that it is *one* recipe that does. A natural question for future work is whether the recipe is unique under specified constraints (e.g., among ratios that are linear in regime-specific binding energy, invoke only the relevant horizon temperature, and require no auxiliary length scale), or whether a family of recipes produces equivalent recurrences. The recurrence observed here is therefore evidence of structural consistency, not yet of structural necessity.

We do not claim that the recurrence reflects a deeper unifying principle; we claim only that it is real, reproducible, and worth naming.

7.3 What the framework is

The framework is a defensible organizing principle that:

- Reproduces known results in three regimes (Bekenstein bound, Bekenstein-Hawking entropy, Casini bound) under a single structural form.
- Identifies the static escrow ratio $|U|/T$ as a recurring structural ratio across distinct sectors of gravitational and entanglement thermodynamics.
- Provides a clear interpretive dictionary connecting matter-side entanglement physics (Casini) to horizon-side entropy (Smarr) to test-mass kinematics (Bekenstein).
- Is anchored by first-principles lattice computation that reproduces the BW identity within lattice-discretization precision in the Rindler-wedge sector.

7.4 Open directions

Several directions remain open for future work:

- **Higher-dimensional lattice verification.** The lattice tests presented here are 1+1-dimensional. Extension to 2+1D and 3+1D would test whether the cross-regime form generalizes to higher dimensions in a non-trivial way; [18] reports preliminary 3+1D modular-Hamiltonian observations indicating that BW recovery is dimension-dependent.
- **Curved-space sectors.** The framework's §2 identification used a Schwarzschild background. [19] extends the horizon recovery to Reissner-Nordström and Kerr via the appropriate Smarr formulas; whether the full recipe-and-evaluation structure of §1.1 survives the inclusion of charge and angular momentum (and what regime-specific (E, L) pairs the recipe extracts in those cases) remains to be worked out in detail.
- **Analog-gravity empirical tests.** [15] derives a complementary non-equilibrium efficiency bound for Bose-Einstein condensate analog gravity, predicting a 17% efficiency suppression in BEC phonon extraction. While that bound arises from Verlinde-based

non-equilibrium thermodynamics rather than from the static escrow recipe of the present work, both belong to the same family of thermodynamic-bound constructions and a unified treatment would strengthen the empirical foundation of the framework.

- **Sharper saturation conditions.** The Escrow Inequality is generally loose. Characterizing the states that saturate the bound (beyond the well-known limit of coherent states modular-aligned with the vacuum) would identify the "Bekenstein-bound saturating" sector of QFT states.

8 Conclusion

We have presented the gravitational entropy escrow framework as a cross-regime organizing observation rather than a derivation. The static escrow recipe $|U|/T$ extracts a regime-specific energy E and length scale L from the gravitational binding energy and horizon temperature; the dimensionless function $\mathcal{N}_{\rm esc}(E, L) = 2\pi EL/(\hbar c)$ — identical in form to the Bekenstein-bound saturation value — is then evaluated at these arguments. Three distinct gravitational regimes (the test-mass sector at the horizon limit via the Smarr partition, the black-hole horizon sector via the Smarr formula, and the localized-perturbation sector via Casini's BW-derivation of the Bekenstein bound on entanglement entropy) all evaluate to the Bekenstein form under this single recipe. The function is Bekenstein's; the recipe is the framework's contribution.

The framework's specific empirical claim is the inequality of Theorem 1 — the escrow form of the Casini–Bisognano–Wichmann bound, with the escrow postulate identifying the right-hand side. Direct lattice computation verifies the inequality across mass-perturbation and Compton-wavepacket protocols; the underlying BW identity is reproduced within lattice-discretization precision ($\sim 0.1\%$) and the bound itself is generally loose (maximum 5.4% saturation observed).

We have been explicit about what the framework does not provide: new derivations from the postulate $|U|/T$, new numerical predictions beyond standard continuum results, or a deeper foundational structure. The contribution is the unifying observation, properly anchored and properly bounded in scope.

Acknowledgments

This manuscript represents v1.3 of a research program developed across multiple iterations and is the fifth deposit in the Windstorm Institute gravitational entropy escrow series (Papers 10–14 in the institute's numbering: [15–19]). Earlier versions of the present synthesis contained claims that did not survive subsequent audit; these have been explicitly retracted in Section 6. The author thanks the adversarial review process — including independent verification runs by separate AI-assisted analysis sessions — for catching errors that would otherwise have appeared in the final manuscript.

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A Lattice computation details

A.1 Free scalar lattice in 1+1D

The discretized Hamiltonian is:

$$H = \frac{1}{2}\sum_{i=1}^{N}\left[\pi_i^2 + (\phi_{i+1} - \phi_i)^2 + m_i^2\phi_i^2\right], \tag{30}$$

with Dirichlet boundary conditions ($\phi_0 = \phi_{N+1} = 0$). In matrix form,

$$H = \frac{1}{2}(\boldsymbol{\pi}^T\boldsymbol{\pi} + \boldsymbol{\phi}^T K\boldsymbol{\phi}), \tag{31}$$

where $K_{ii} = 2 + m_i^2$, $K_{i,i\pm1} = -1$.

Diagonalizing $K = U\Omega^2U^T$ with $\Omega = \mathrm{diag}(\omega_k)$, the vacuum covariance matrices are:

$$X_{ij} = \langle\phi_i\phi_j\rangle_{\mathrm{vac}} = \frac{1}{2}(U\Omega^{-1}U^T)_{ij}, \quad P_{ij} = \langle\pi_i\pi_j\rangle_{\mathrm{vac}} = \frac{1}{2}(U\Omega U^T)_{ij}. \tag{32}$$

A.2 Entanglement entropy via Williamson decomposition

For a subsystem A with covariance matrices X_A, P_A, the symplectic eigenvalues ν_k are obtained from:

$$\nu_k^2 = \mathrm{eig}(X_A P_A). \tag{33}$$

The von Neumann entropy is:

$$S = \sum_k\left[(\nu_k + 1/2)\log(\nu_k + 1/2) - (\nu_k - 1/2)\log(\nu_k - 1/2)\right], \tag{34}$$

with the convention that only $\nu_k > 1/2 + \epsilon$ contribute.

A.3 Modular Hamiltonian and boost generator

The lattice modular Hamiltonian M_K acts on the doubled (position+momentum) phase space:

$$M_K = i\Omega\log\left[\frac{\gamma + i\Omega/2}{\gamma - i\Omega/2}\right], \tag{35}$$

where $\gamma = \mathrm{diag}(X_A, P_A)$ and Ω is the symplectic form.

The boost generator is computed directly as:

$$\langle K_A^{\mathrm{boost}}\rangle = 2\pi\sum_{i\in A}(i - \mathrm{cut})\cdot\langle T_{00}(i)\rangle, \tag{36}$$

with the lattice energy density:

$$T_{00}(i) = \frac{1}{2}\left[\langle\pi_i^2\rangle + \langle(\phi_{i+1} - \phi_i)^2\rangle + m_i^2\langle\phi_i^2\rangle\right]. \tag{37}$$

A.4 Wavepacket construction

For Protocol A, a Gaussian wavepacket of width $\sigma_c = 1/m$ centered at site $i_0 = \mathrm{cut} + d_1$ is:

$$f_i = \frac{1}{\mathcal{N}}\exp\left[-\frac{(i - i_0)^2}{4\sigma_c^2}\right], \quad \sum_i |f_i|^2 = 1. \tag{38}$$

The perturbed covariances are:

$$\delta X_{ij} = 2(Gf)_i(Gf)_j, \quad \delta P_{ij} = 2(Hf)_i(Hf)_j, \tag{39}$$

where $G = U\Omega^{-1/2}U^T/\sqrt{2}$ and $H = U\Omega^{1/2}U^T/\sqrt{2}$ are the mode functions.

A.5 Numerical precision

All computations performed in double precision. Williamson decomposition uses standard linear algebra routines (LAPACK `geev`, `eigh`). Logarithm of matrices uses Schur decomposition (`scipy.linalg.logm`). Reported precision in tables reflects measurement-to-measurement reproducibility, not absolute numerical accuracy.